

30066- MACROECONOMICS (8 CFU)

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(Ch. 3)

The Goods Market: exercises and
practice

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USING THE MODEL: THE PARADOX OF THRIFT

Suppose that the geopolitical instability increases future uncertainty and consumers want to increase their saving , for any income level

Negative demand Shock: $c_0 \downarrow \Rightarrow \Delta c_0 < 0$

What is the impact of this shock on Y , C and S ?

IMPACT ON Y

Quantify the change in Y starting from its equilibrium level by applying the Δ to both members

$$Y = \frac{1}{1 - c_1} (c_0 - c_1 T + \bar{I} + G)$$

$$\Delta Y = \frac{1}{1 - c_1} (\Delta c_0 - c_1 \Delta T + \Delta \bar{I} + \Delta G)$$

$$\Delta Y = \frac{1}{1 - c_1} (\Delta c_0 - c_1 0 + 0 + 0)$$

$$\Delta Y = \frac{1}{1 - c_1} \Delta c_0$$

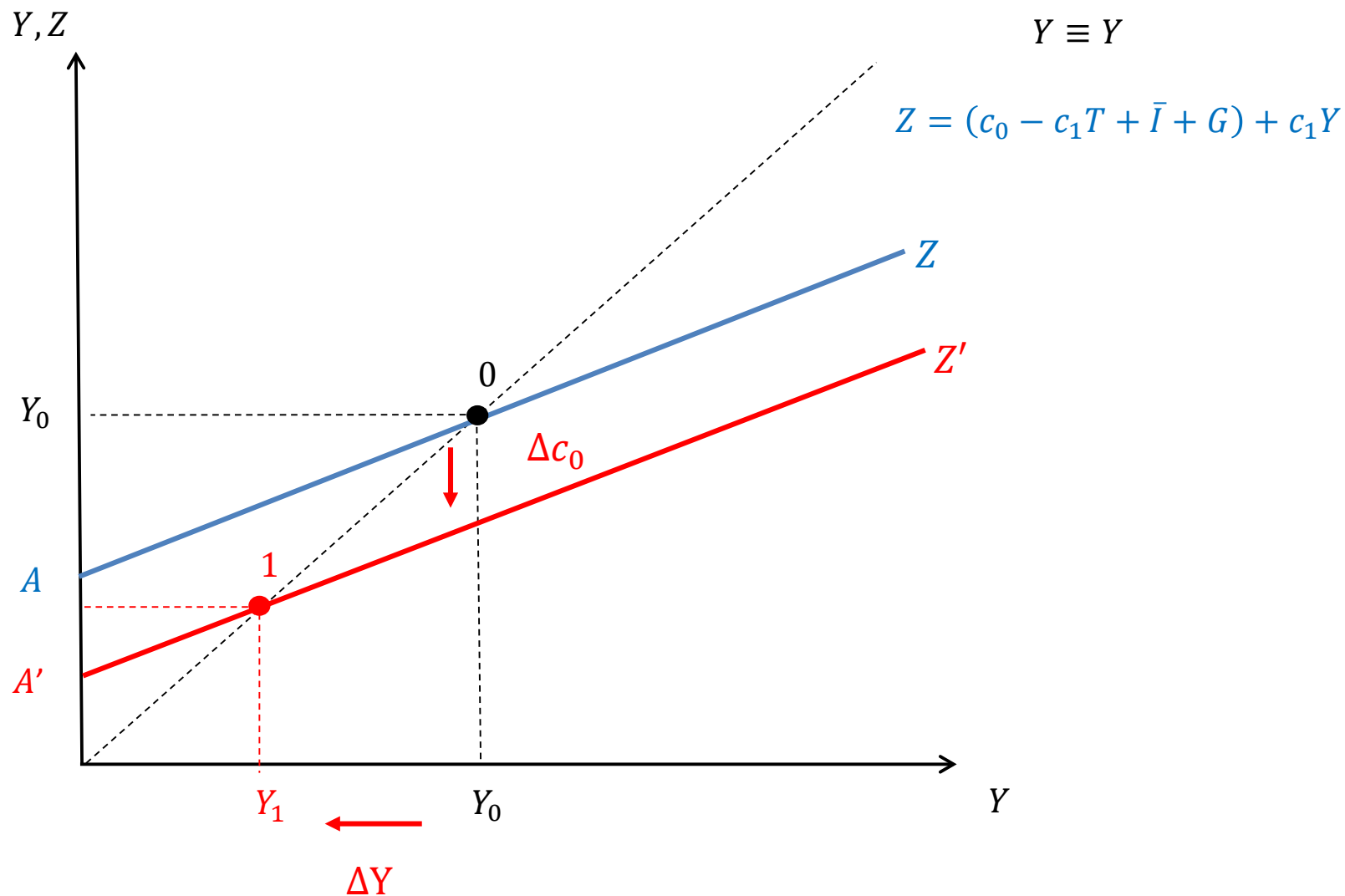
- If $c_0 \downarrow \Rightarrow Y \downarrow$

...but the impact is more than proportional

$$|\Delta Y| > |\Delta c_0|$$

=> There is a multiplier effect

GRAPHICAL ANALYSIS



INTUITION

When c_0 changes, Y changes because of two mechanisms

- **Direct effect:** $c_0 \downarrow \Rightarrow Z \downarrow \Rightarrow Y \downarrow$
 - The exogenous reduction of consumption reduces demand, hence production and income
- **Indirect effect:** $\Rightarrow Y \downarrow \Rightarrow C \downarrow \Rightarrow Z \downarrow \Rightarrow Y \downarrow \Rightarrow C \downarrow \Rightarrow Z \downarrow \dots \Rightarrow Z \downarrow \downarrow$
 - The reduction in income leads to a further decline in consumption and demand, reducing production and income more than proportionally

IMPACT ON CONSUMPTION

Quantify the change in C starting from its equation by applying the Δ to both members and remembering that c_0 changes **but, in equilibrium, also Y changes**

$$C = c_0 + c_1 Y - c_1 T$$

$$\Delta C = \Delta c_0 + c_1 \Delta Y - c_1 \Delta T$$

$$\Delta C = \Delta c_0 + c_1 \Delta Y - c_1 0$$

$$\Delta C = \Delta c_0 + c_1 \frac{1}{1 - c_1} \Delta c_0$$

$$\Delta C = \frac{1 - c_1 + c_1}{1 - c_1} \Delta c_0$$

$$\Delta C = \frac{1}{1 - c_1} \Delta c_0 = \Delta Y$$

C decreases for two reasons:

1. $c_0 \downarrow \Rightarrow C \downarrow$
2. in equilibrium $Y \downarrow, C \downarrow$

- Consumption declines more than proportionally compared to the initial exogenous reduction
- The decline in consumption is exactly equal to the decline in income

IMPACT ON SAVING

Quantify the change in S starting from its equation by applying the Δ to both members and remembering that c_0 changes **but, in equilibrium, also Y changes**

$$S = -c_0 + (1 - c_1)(Y - T)$$

$$\Delta S = -\Delta c_0 + (1 - c_1)(\Delta Y - \Delta T)$$

$$\Delta S = -\Delta c_0 + (1 - c_1)(\Delta Y - 0)$$

$$\Delta S = -\Delta c_0 + (1 - c_1) \frac{1}{1 - c_1} \Delta c_0$$

$$\Delta S = -\Delta c_0 + \Delta c_0 = 0$$

Paradox of saving: we wanted to save more but in equilibrium aggregate private saving does not change!

- We can prove the previous result by considering the alternative equilibrium condition:

$$S_{\text{Nat}} = I$$

$$S_{\text{Priv}} + S_{\text{Gov}} = I$$

Since S_G does not change and $\bar{I}$ does not change in equilibrium also the private saving stays constant

PARADOX SAVING:

If the goods market is in equilibrium, and I is exogenous a change in autonomous consumption does not affect nor private nor public saving

EXTENSIONS

Assume now that investment is **endogenous**, i.e. it depends on **income**

$$I = I_{+}(Y) = \bar{I} + d_1 Y$$

Is the impact of a reduction of c_0 different?
Does the paradox still hold?

Aggregate demand: $Z = C + I + G$

$$Z = c_0 - c_1(Y - T) + \bar{I} + d_1Y + G$$

$$Z = (c_0 - c_1T + \bar{I} + G) + (c_1 + d_1)Y$$

Find the equilibrium:

$$\begin{cases} Y = Z \\ Z = (c_0 - c_1T + \bar{I} + G) + (c_1 + d_1)Y \end{cases}$$

$$Y = (c_0 - c_1T + \bar{I} + G) + (c_1 + d_1)Y$$

$$(1 - c_1 - d_1)Y = (c_0 - c_1T + \bar{I} + G)$$

$$Y = \frac{1}{(1 - c_1 - d_1)} (c_0 - c_1T + \bar{I} + G)$$

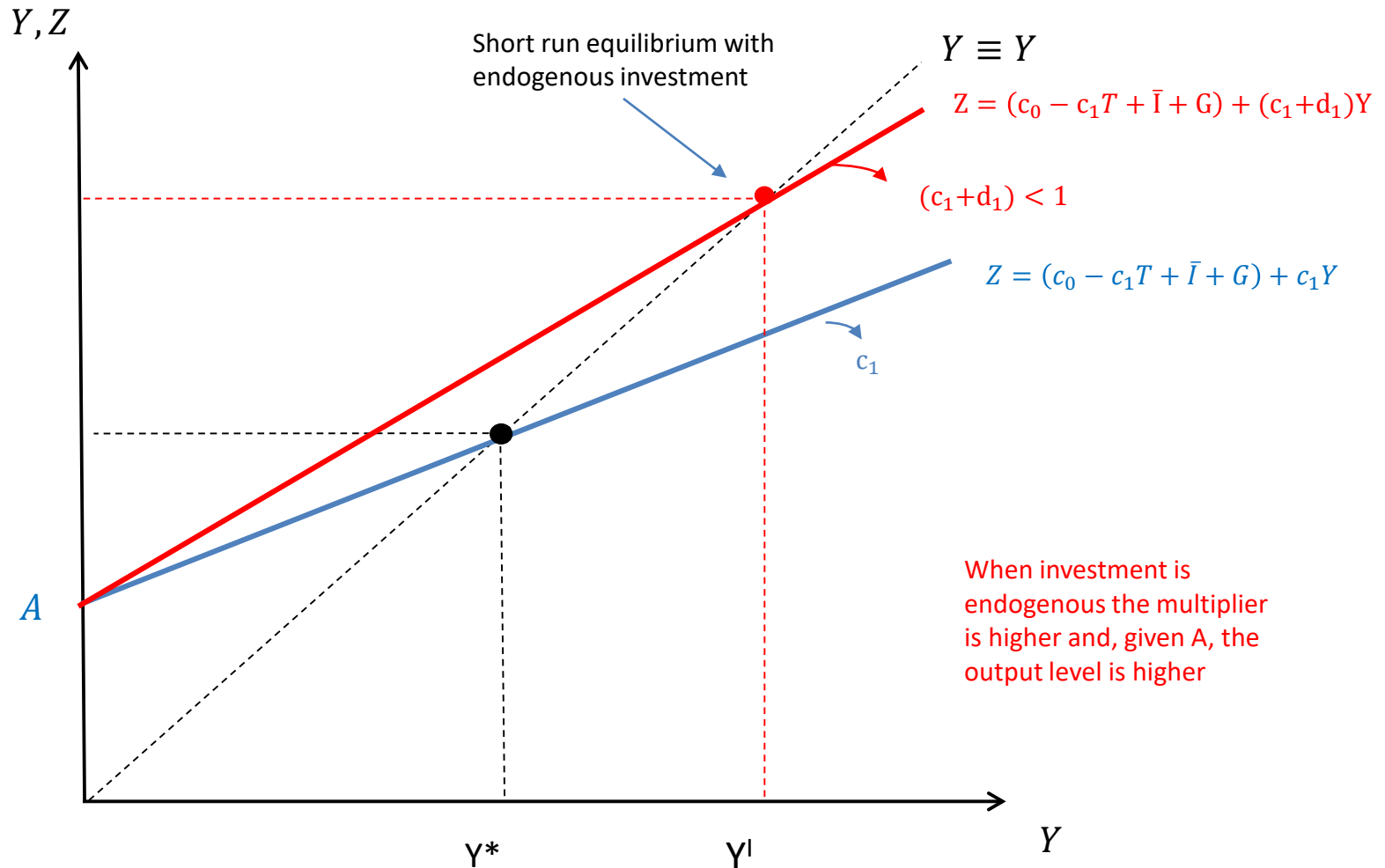


Higher multiplier compared to the standard case: two channels (C and I) that stimulate indirectly demand and income



Same autonomous spending as in the standard case

GRAPHICAL ANALYSIS



- If c_0 decreases, aggregate demand decreases and, through the multiplier, income decreases more than proportionally
- In equilibrium investment will be lower
- Using the equilibrium condition between saving and investment

$$S_{\text{Nat}} = I$$

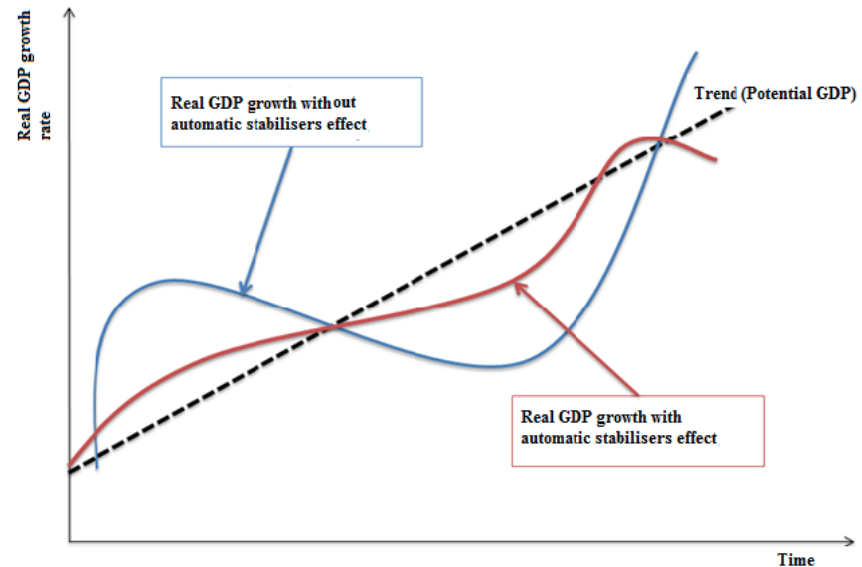
$$S_{\text{Priv}} + S_{\text{Gov}} = I$$

S_G does not change

I decreases

To satisfy the equality, also private saving have to decrease

The PARADOX SAVING does not hold in this case



APPLICATION: AUTOMATIC STABILIZERS - ENDOGENOUS TAXATION

- Suppose that **taxes are endogenous**:

$$T = t_0 + t_1 Y \quad 0 < t_1 < 1$$


Tax rate

- All other behavioral equations are the standard ones
-
- a) Solve for the equilibrium
 - b) Compare the multiplier with the one of the standard model
 - c) Why does endogenous taxation represent an “automatic stabilizer” for the economy?

a) EQUILIBRIUM

Find demand with the specific assumptions

$$Z = C + I + G$$

$$C = c_0 + c_1 Y - c_1 T \qquad T = t_0 + t_1 Y$$

Substitute T into consumption

$$C = c_0 + c_1 Y - c_1(t_0 + t_1 Y)$$

$$C = c_0 + c_1 Y - c_1 t_0 - c_1 t_1 Y$$

$$C = c_0 - c_1 t_0 + c_1(1 - t_1)Y$$

Substitute into Z

$$Z = C + I + G$$

$$Z = c_0 - c_1 t_0 + c_1(1 - t_1)Y + \bar{I} + G$$

$$Z = (c_0 - c_1 t_0 + \bar{I} + G) + c_1(1 - t_1)Y$$

Aggregate demand line is flatter compared to the standard case

Impose the equilibrium condition using the new Z function and solve for Y:

$$\begin{cases} Y = Z \\ Y = (c_0 - c_1 t_0 + \bar{I} + G) + c_1(1 - t_1)Y \end{cases}$$

$$Y = A + c_1(1 - t_1)Y$$

$$Y - c_1(1 - t_1)Y = [1 - c_1(1 - t_1)]Y = A$$

$$Y = \frac{1}{1 - c_1(1 - t_1)} A$$

b) DIFFERENCES w.r.t. STANDARD CASE

Standard case

$$\frac{1}{1 - c_1}$$

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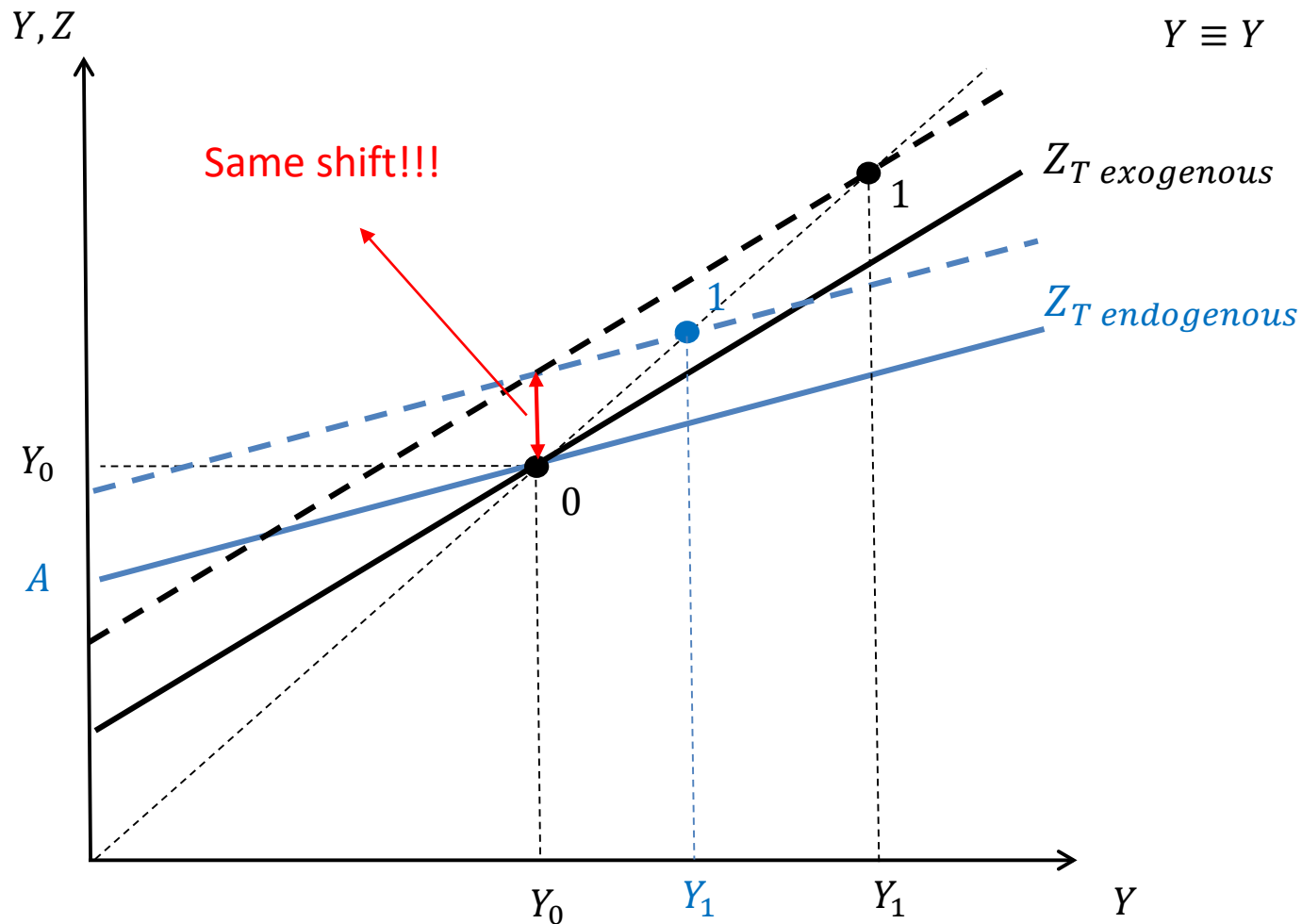
Endogenous taxes

$$\frac{1}{1 - c_1(1 - t_1)}$$

Compared with the standard case the demand function is flatter and the multiplier is smaller:

Lower impact on Y of any changes in autonomous spending

GRAPHICAL ANALYSIS



c) WHY THIS TAXATION RULE IS AN «AUTOMATIC STABILIZER»?

- The impact on Y of any changes in autonomous spending is **smaller**
 - When income increases, taxes increase too, reducing the positive impact of income on consumption!!!
 - **Lower final impact on income Y**
 - Avoid wide fluctuations in Y
- $A \uparrow \Rightarrow Z \uparrow \Rightarrow Y \uparrow$
 - $C \uparrow \Rightarrow Z \uparrow \Rightarrow Y \uparrow$
 - $T \uparrow \Rightarrow C \downarrow \Rightarrow Z \downarrow \Rightarrow Y \downarrow$

Mitigation effect on Y :
 Y increases less than with exogenous
taxation

**APPLICATION:
FISCAL POLICY WITH A BALANCED
BUDGET**

INTRODUCTION

- **AIM:** The Government wants to stimulate the economy but keeping balanced the public budget
- Hence, it proposes a simultaneous increase, by exactly the same amount, of both government expenditure and taxes, i.e.

$$\Delta T = \Delta G > 0$$

- The opposition party argues that, by increasing taxes exactly by the same amount as expenditure, the fiscal policy will have **NO impact on the economy** (no stimulus). The government disagrees.

Which of the two is right? The government or the opposition party?

ANALYTICAL SOLUTION

Quantify the change in Y starting from its equilibrium level by applying the Δ to both members

$$Y = \frac{1}{1 - c_1} (c_0 - c_1 T + \bar{I} + G)$$

$$\Delta Y = \frac{1}{1 - c_1} (\Delta c_0 - c_1 \Delta T + \Delta \bar{I} + \Delta G)$$

$$\Delta Y = \frac{1}{1 - c_1} (0 - c_1 \Delta T + 0 + \Delta G)$$

$$\Delta Y = \frac{1}{1 - c_1} (-c_1 \Delta T + \Delta G)$$

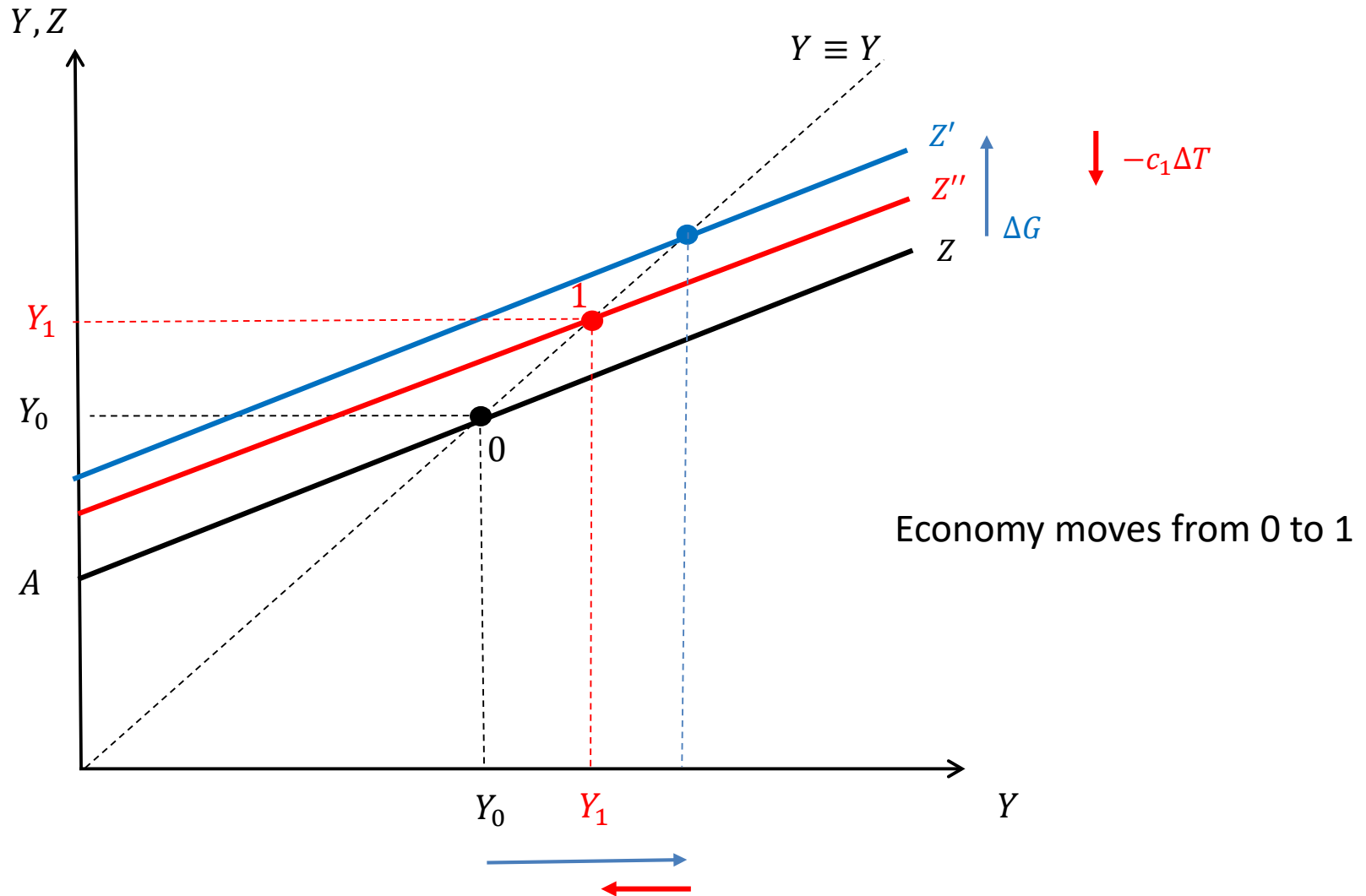
$$\Delta Y = \frac{1}{1 - c_1} (-c_1 \Delta T + \Delta T) = \frac{1 - c_1}{1 - c_1} \Delta T$$

$$\Delta Y = \Delta T > 0$$

**GDP
increases!!!**

GRAPHICAL ANALYSIS

$$Z = (c_0 - c_1T + \bar{I} + G) + c_1Y$$



INTUITION

- Impact of $G \uparrow$: $G \uparrow \Rightarrow Z \uparrow$ proportionally
 - If $\Delta G = 1\text{€}$, the initial impact on Z is exactly 1€
 - Impact reinforced by the multiplier
- Impact of $T \uparrow$: $T \uparrow \Rightarrow C \downarrow \Rightarrow Z \downarrow$ less than proportionally
 - The impact of taxes is mediated by the marginal propensity to consume
 - If $\Delta T = 1\text{€}$, the initial impact on Z is $-c_1\text{€}$
 - Impact reinforced by the multiplier

The sum of the two shocks increases the demand by $(1 - c_1)\text{€}$
and through the multiplier, the production increases by the same amount of
the change in G or T